Publication and IPR Plan

Where to publish the project, in what order, and what to decide first

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Reading guide. This plan separates what was verified from what was not. Dates and rules that were read on a call-for-papers page are marked *read*; everything else is marked *unverified* and must be confirmed by the authors on the venue's own page before any commitment. Nothing here is legal advice; patent matters need a registered patent professional.

1. Recommendation in one page

1. **Decide the IPR question first (this week).** Publishing the paper or pushing the code makes the work public. In many jurisdictions public disclosure before filing destroys novelty. The institution's IPR cell, the guide, or a patent professional should decide whether any element is worth filing. The invention disclosure draft (`08_IPR_Invention_Disclosure_Draft`) is ready for that conversation. Until the decision, do not push to GitHub and do not post a preprint.
2. **If no filing is wanted: submit the paper to IEEE/ACM CCGrid 2027** (double-blind, full paper up to 10 pages including references, IEEE two-column format; abstract due 24 Nov. 2026 and paper due 1 Dec. 2026, as *read* on the call page; re-check). The anonymous Lab-10 draft `06b` has 14 pages; CCGrid allows 10 pages including references, so a 10-page conference cut (drop the appendices, the earlier-study details and the long tables) must be made first. Weakness the reviewers will raise: one machine, one workload family, a hand rule within 1% of the dispatcher, emulated remote. State these limits in the paper (the draft does).
3. **If the Raspberry Pi result is wanted first:** target a journal (Future Generation Computer Systems, Journal of Parallel and Distributed Computing, IEEE Transactions on Parallel and Distributed Systems, or Journal of Systems Architecture, all *unverified* for current scope, fees and turnaround) with an extended version that includes real-device results and the second workload family.
4. **Book chapter:** only if a specific call exists (*unverified*; none was found in this work). Do not select one by guess.
5. **Whatever the route:** fill author, institution, guide and e-mail placeholders; confirm authorship order and the guide's consent; check that the institution permits publication of the project.

Table 1. Route comparison

Route	Fit	Effort before submission	Main risk	Status of facts
CCGrid 2027 (full paper)	high: cluster, cloud and grid scheduling; empirical systems papers; double-blind	low: 06b ready; polish and anonymise artefacts	reviewers want real remote hardware and a second workload; tight deadline	dates read on the call page; confirm
Workshop co-located with a major systems conference	medium: lower bar, shorter paper	low to medium	workshop list and dates unknown	unverified

Route	Fit	Effort before submission	Main risk	Status of facts
Journal (extended version, after Pi measurement)	high: room for 12+ pages, statistics, appendices	high: Pi runs, second workload, clean-tree rerun	longer review; fees at some journals	unverified; check scope pages
Book chapter	medium: survey-style content (Review 1) suits it	medium	no verified call; publisher terms	unverified
Patent / IPR (India or US)	low to medium: base patent and IBM grant occupy the obvious ideas	medium: IDF, prior-art search	publication-before-filing bar; cost; weak claims	needs professional advice

2. Conference route: CCGrid 2027

2.1 What was read

The call for papers at [1] was read in November 2026. It specifies IEEE two-column format, a limit of 10 pages including references for full papers, double-blind review, an abstract deadline of 24 Nov. 2026 and a paper deadline of 1 Dec. 2026. These dates were captured on the day of reading; call pages are updated, so **re-open the page and confirm the dates, page limits, artifact-evaluation rules and anonymity rules before submitting.**

2.2 Fit and framing

- Fit: scheduling of workflows on heterogeneous resources, edge-cloud continuum, performance evaluation, reproducibility.
- Frame the contribution as a measured, overhead-accounted study of when prediction helps, with a proposed dispatcher, not as a claim of a new learning algorithm. This matches the evidence and avoids the IBM-grant overlap.
- Lead with the final benchmark: with one cost model on both sides the dynamic dispatcher beats a static HEFT plan by 5.2% (CPU) and 12.2% (strict GPU set) and blind dispatch by 17.8% and 21.2%; then give the limits (a hand rule within 1%, a tie without cost spread, 23% to 31% GPU model error, an earlier session that found HEFT level).
- Anonymity: use `06b`; the repository link must be replaced by an anonymised artefact link or a statement; remove names from file metadata, figure captions and the dashboard screenshot.

2.3 Checklist before 24 Nov. 2026

1. Authors, affiliations, guide permission and e-mail in the non-anonymous version; confirm the IPR decision.
2. Re-read the call page; confirm abstract and paper dates, page limit and double-blind rules.
3. Regenerate figures only from stored result files; confirm every number against `RESULTS_TABLE.csv` and `claims.json`.
4. Decide whether to run a quick second workload family or the Pi before 1 Dec.; if not, keep the limitations paragraph as is.
5. Prepare an anonymised artefact (zip of the reproducibility subset) if the call allows an artefact link.
6. Register the abstract by 24 Nov., upload the PDF by 1 Dec.

2.4 Realistic reviewer comments and answers

Table 2. Anticipated reviewer comments

Comment	Answer available in the paper
Gain is not larger than an informed rule	Agreed and stated; the contribution is a generic learned model, overhead accounting and a map of when it helps
Workload is designed to favour the method	Stated as a threat; Montage, uniform and two-factor nulls are reported; dose and heavy-count sweeps show where it fails
No real remote device	Labelled emulation; runbook and rehearsal; Pi measurement planned
Only one machine	Stated; environment hash recorded; results never pooled across machines
No comparison with learned DRL schedulers	Not run; stated; surveyed gains are simulation-only and not comparable
Sessions have different code hashes and a dirty tree	Disclosed; never pooled; sprint changes do not touch the scheduler path

3. Journal and workshop routes

These are candidates based on where the surveyed papers appeared and on venue scope. Impact factors, fees, scope statements and turnaround times were **not verified** in this work; check each journal's own page.

Table 3. Candidate journals

Journal	Why a candidate	Evidence in this work
Future Generation Computer Systems	published workflow-runtime prediction ([2]) and edge-offloading DRL work ([3])	appears in the survey
Journal of Systems Architecture	published lightweight heterogeneous scheduling on a real board ([4])	appears in the survey
IEEE Transactions on Parallel and Distributed Systems	published HEFT and PEFT ([5], [6])	appears in the references
Concurrency and Computation: Practice and Experience	published the StarPU journal paper ([7])	appears in the references
Journal of Cloud Computing	published several offloading papers ([8], [9])	appears in the survey

Table 4. Indexing and quartile status of the candidates (checked 3 Nov. 2026 on aggregator pages that quote SCImago; the quartile is category-specific and changes yearly, so confirm on scimagojr.com and Scopus before choosing)

Venue	Reported quartile (SJR best quartile)	How certain
IEEE/ACM CCGrid 2027 (conference)	conference; IEEE proceedings; Scopus indexing of the 2027 volume not yet verifiable	dates verified on the call page: abstract 24 Nov. 2026, paper 1 Dec. 2026 (AoE), notification 1 Feb. 2027, camera-ready 16 Mar. 2027; 10 pages including references; double-blind; conference 24 to 27 May 2027, Dallas Fort-Worth
Future Generation Computer Systems	Q1 (several computer science categories)	aggregators agree; SJR value varies by source
IEEE Trans. Parallel and Distributed Systems	Q1	aggregators agree; SJR value varies by source

Venue	Reported quartile (SJR best quartile)	How certain
Journal of Systems Architecture	Q1 (Hardware and Architecture; Software)	one aggregator; confirm
Journal of Parallel and Distributed Computing	Q1 (several categories)	one aggregator; confirm
Concurrency and Computation: Practice and Experience	mixed: CiteScore Q1 or Q2 by category, JCR Q3	no clear SJR quartile found; weakest candidate

Recommended order. (1) CCGrid 2027 with a 10-page cut of the Lab-10 paper if the IPR decision allows public submission; (2) otherwise or afterwards a Q1 journal (Future Generation Computer Systems or IEEE TPDS) with the extended version and real-device results. A book chapter needs a specific call (none verified).

A journal paper should add: real Pi results, a second workload family, a rerun of the headline experiment on a clean committed tree, and the dispatch-loop fix applied in the source. That makes the claim of generality testable and removes three of the four threats a reviewer will raise.

4. IPR route

4.1 What the prior art already covers

- Task graphs, eligibility-gated queues, in-degree counters and utilisation-based scaling: claimed by US 11,561,826 B1 ([10]).
- Learned rank assigned to tasks of a DAG and rank-ordered execution on a heterogeneous SoC: disclosed by IBM US 11,740,933 B2 ([11]); task execution estimates in IBM US 11,966,776 B2 ([12]).
- Per-run profiling for CPU+GPU work split: Intel US 10,186,007 B2 ([13]); duration prediction under core contention: Samsung US 10,162,671 B2 ([14]), lapsed.
- Runtime performance models for CPU/GPU DAG scheduling: StarPU ([7]), published; list scheduling: HEFT and PEFT ([5], [6]), published.

4.2 Candidate elements and their weaknesses

Table 5. Candidate inventive elements (not claims; for discussion with a professional)

Element	Possible novelty	Main weakness
A: a late-binding dispatcher that orders ready tasks by predicted upward rank using a model fitted per (stage, pool) from execution logs on whichever numeric parameters vary, with no stage- or parameter-specific code	generic parameter discovery from logs; refusal for unseen stage/pool; one-command refit for a new pool layout	IBM rank-ordered learned dispatch; HEFT-type upward rank; ridge regression is standard; the result equals an informed rule
B: a pre-deployment headroom gate: replay logs with an oracle estimator and enable predictive scheduling only if the headroom exceeds a threshold (5% in our design)	decision method for when prediction is worth its cost	method of evaluation may be treated as an abstract idea; weak technical effect; one gate failure shown (V1)
C: two-metric overhead accounting (per-dispatch decision time versus coordinator scheduling time) with a detector for an idle ineligible worker defeating the dispatch loop's early exit	diagnosis and fix of a dispatch-loop inefficiency	software engineering detail; small effect (under one point on the gain)
D: transfer-aware placement with remote-worker predictions	avoids harm of blind remote placement	no offloading gain was established; emulation only

Assessment. On the evidence read, element A is the only one with a plausible technical effect, and it sits close to the IBM grant; a patent on it would be narrow and uncertain. The measured result (level with an informed rule) weakens an argument of non-obviousness. A copyright/open-source release of the code and a publication are likely to deliver more value than a patent. The decision belongs to the authors, the guide and the institution.

4.3 Timing rules

1. Do not publish a paper, preprint, poster, public repository, talk slide or demo video describing elements A to D before a filing decision is made. Public disclosure may bar patentability in many jurisdictions; grace periods differ by country and are not safe to rely on.
2. If filing is wanted, give the invention disclosure draft to the institution's IPR cell or a patent agent, commission a prior-art search, and file (a provisional, where available) before submitting the paper.
3. A paper under double-blind review is not yet public, but its content may be disclosed to reviewers; ask the professional whether filing before submission is needed.
4. Keep the authors' contribution records (who conceived which element, with dates); the git history and ACTIVITY_LOG.md help.

4.4 Software licence

Choosing a licence is left to the authors: the repository contains third-party material (Montage, DSS2 images) with their own terms. For the GitHub package, exclude large or unlicensed data; choose a licence only after the IPR decision and check the institution's rules on ownership.

5. Decisions needed from the authors

Table 6. Decision list

#	Decision	Default if undecided
1	File any IPR? (IDF draft ready)	do not push, do not post a preprint; submit nothing public
2	Submit to CCGrid 2027 by 24 Nov. / 1 Dec. 2026?	no submission; keep 06b ready
3	Run the Pi and second workload for a journal version?	journal route deferred
4	Authors' names, order, institution, guide, e-mail	placeholders remain
5	Repository licence and visibility	private, unpushed

References

[1] CCGrid 2027 Call for Papers, https://easychair.org/cfp/ccgrid27 (accessed Nov. 2026).

[2] J. Bader, F. Lehmann, L. Thamsen, U. Leser, and O. Kao, "Lotaru: Locally predicting workflow task runtimes for resource management on heterogeneous infrastructures," *Future Gener. Comput. Syst.*, vol. 150, pp. 171–185, 2024, doi: 10.1016/j.future.2023.08.022.

[3] J. Liu, Y. Mi, X. Zhang, and X. Li, "Task graph offloading via deep reinforcement learning in mobile edge computing," *Future Gener. Comput. Syst.*, vol. 158, pp. 545–555, 2024, doi: 10.1016/j.future.2024.04.034 (arXiv:2309.10569).

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[6] H. Arabnejad and J. G. Barbosa, "List scheduling algorithm for heterogeneous systems by an optimistic cost table," *IEEE Trans. Parallel Distrib. Syst.*, vol. 25, no. 3, pp. 682–694, Mar. 2014, doi: 10.1109/TPDS.2013.57.

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[10] S. Nagpal, A. Karumannil, V. Jain, and A. K. Patil (Xilinx, Inc.), "Scheduling processing of machine learning tasks on heterogeneous compute circuits," U.S. Patent 11,561,826 B1, Jan. 24, 2023 (filed Nov. 12, 2020).

[11] A. Vega *et al.* (International Business Machines Corp.), "Heterogeneous system on a chip scheduler with learning agent," U.S. Patent 11,740,933 B2, Aug. 29, 2023.

[12] A. Amarnath *et al.* (International Business Machines Corp.), "Learning agent based application scheduling," U.S. Patent 11,966,776 B2, Apr. 23, 2024.

[13] R. Barik, T. Shpeisman, B. T. Lewis, and R. Kaleem (Intel Corp.), "Adaptive scheduling for task assignment among heterogeneous processor cores," U.S. Patent 10,186,007 B2, Jan. 22, 2019.

[14] R. Kim and M.-S. Kim (Samsung Electronics Co., Ltd.), "Method and apparatus for performing task scheduling in terminal," U.S. Patent 10,162,671 B2, Dec. 25, 2018.